

$$\begin{aligned}
 \mathbb{E}[(T-a)\psi(T)] &= 0 \quad \psi(T) = \begin{cases} 1 & \text{if } T < c_1 \text{ or } T > c_2 \\ 0 & \text{otherwise} \end{cases} \quad \frac{26/10/2024}{\text{Para Inf}} \\
 \left(\underbrace{T-a}_X \stackrel{d}{=} -(T-a) \right) &= \mathbb{1}(T-a < \underbrace{c_1-a}_{K_1} \text{ or } T-a > \underbrace{c_2-a}_{K_2}) \\
 &= \mathbb{1}(X < K_1 \text{ or } X > K_2) \\
 &\quad \text{where } K_1 = -K_2 \\
 &= \mathbb{E}[X \mathbb{1}(|X| > K_2)] \\
 &= \int_{-\infty}^{\infty} x \mathbb{1}(|x| > K_2) f_X(x) dx \quad (f_X(x) = f_X(-x)) \\
 &= \mathbb{E}X - \int_{-K_2}^{K_2} x f_X(x) dx = 0
 \end{aligned}$$

p-Value H_0 vs. $H_1 \dots \underline{X}$. output: $\underset{0}{\text{accept}} / \underset{1}{\text{reject}} H_0$

Facts

- ① p-Value is a function of data (statistic), denoted as $p(\underline{x})$ ($p(\underline{x}) \in [0, 1]$) which measures the strength of evidence against H_0
- ② Smaller the $p(\underline{x})$, stronger the evidence we have against H_0 .

For a level- α test: we need to reject H_0 if $p(\underline{x}) \leq \alpha$
 should have $\mathbb{P}_{H_0}(p(\underline{X}) \leq \alpha) \leq \alpha$

Testing: $H_0: \theta \in \odot_0$ vs. $H_1: \theta \in \odot_1$, based on some statistic $T(X)$

Defⁿ 1: p-value is the probability that the test statistic $T(X)$ is as extreme (in the direction of H_1) as the observed test stat $T(\underline{x})$ (or T_{obs}) when H_0 is true.

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2)$
 $\searrow$ known

Find the p-value for testing $H_0: \mu = \mu_0$

vs. $H_1: \mu > \mu_0$

using $T(X) = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0}$

$$p(\underline{x}) = \mathbb{P}_{\mu_0} \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} \right)$$

$$= 1 - \Phi \left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} \right)$$

extremeness is measured w.r.t. the null distribution of $T(X)$

Ex. Find p-value for testing $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$

based on $T(X) = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0}$ extreme $\Leftrightarrow$ large +ve or -ve value of $T(X)$

Then $p(\underline{x}) = \mathbb{P}_{\mu_0} (T(X) > |T_{obs}| \text{ or } < -|T_{obs}|)$

$$= 2 \mathbb{P}_{\mu_0} (T(X) > |T_{obs}|)$$

$$= 2(1 - \Phi(|T(\underline{x})|))$$

Drawback: $H_0: \mu \leq \mu_0$ vs. $H_1: \mu > \mu_0$

$$p_{\mu}(\underline{x}) = \mathbb{P}_{\mu} \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0} > \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} \right) \text{ for } \mu \leq \mu_0$$

Define $p(\underline{x}) := \sup_{\mu \leq \mu_0} p_{\mu}(\underline{x})$

$X \leq_{st} Y$ (Y is "stochastically larger" than X)

if $\mathbb{P}[Y > y] \geq \mathbb{P}[X > y]$ i.e. $F_Y(y) \leq F_X(y) + y$

$$p(x) := \sup_{\mu \leq \mu_0} p_{\mu}(x) \quad \text{We need: } \mathbb{P}_{\mu}[p(X) \leq u] \leq u \quad \forall 0 \leq u \leq 1$$

$$\downarrow$$

$$\text{Show } \mathbb{P}_{\mu}(p(X) \leq \alpha) \leq \alpha \quad \forall \mu \leq \mu_0$$

(Given H_0, H_1 , rejection region R)

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2) \quad H_0: \mu = \mu_0 \text{ vs. } H_1: \mu > \mu_0$

$$T(x) = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma_0}$$

$$R_{\alpha} = \left\{ \underline{x} : \underbrace{\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}}_{T_{obs}} > z_{\alpha} \right\} \quad R_{\alpha_1} \subseteq R_{\alpha_2}$$

$$\alpha_1 < \alpha_2$$

A sequence of rejection region is called "nested" if $R_{\alpha_1} \subseteq R_{\alpha_2} \quad \forall \alpha_1 < \alpha_2$

Def 2: (Given H_0, H_1, R_{α})

If the rejection R_{α} for testing H_0 vs. H_1 is nested, then we define p-value $p(\underline{x}) := \inf \{ 0 < \alpha < 1 : \underline{x} \in R_{\alpha} \}$

Ex $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2) \quad H_0: \mu = \mu_0 \text{ vs. } H_1: \mu > \mu_0$

$$R_{\alpha} = \left\{ \underline{x} : \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_{\alpha} \right\}$$

$$= \left\{ \underline{x} : \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) > \underbrace{\Phi(z_{\alpha})}_{1-\alpha} \right\}$$

$$= \left\{ \underline{x} : \alpha > 1 - \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) \right\}$$

$$\underline{x} \in R_{\alpha} \Leftrightarrow \alpha > 1 - \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) \therefore p(\underline{x}) = 1 - \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right)$$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2)$

$\hookrightarrow$ known

Find the p-value of the test

$$\phi(\underline{x}) = \mathbb{1}\left(\frac{\sqrt{n}|\bar{x} - \mu_0|}{\sigma_0} > z_{\alpha/2}\right)$$

for testing $H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$

$$R_\alpha = \left\{ \underline{x} : \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0} > z_{\alpha/2} \text{ or } < -z_{\alpha/2} \right\}$$

$$= \left\{ \underline{x} : \underbrace{\Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right)}_{1-\alpha/2} > \underbrace{\Phi(z_{\alpha/2})}_{\alpha/2} \text{ or } < \underbrace{\Phi(-z_{\alpha/2})}_{\alpha/2} \right\}$$

$$= \left\{ \underline{x} : \alpha > 1 - \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) \text{ or } \alpha > \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) \right\}$$

$$= \left\{ \underline{x} : \alpha > 2 \min \left\{ \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right), 1 - \Phi\left(\frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma_0}\right) \right\} \right\}$$

$$\therefore p(\underline{x}) = 2(1 - \Phi(|T_{obs.}|))$$

Ex. $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$ Test $H_0: \sigma = \sigma_0$ vs. $H_1: \sigma > \sigma_0$. Using

test $\phi(\underline{x}) = \mathbb{1}\left(\frac{\sum x_i^2}{\sigma_0^2} > \chi_{n, \alpha}^2\right)$

$$R_\alpha = \left\{ \underline{x} : \frac{\sum x_i^2}{\sigma_0^2} > \chi_{n, \alpha}^2 \right\}$$

$$= \left\{ \underline{x} : F_{\chi_n^2}\left(\frac{\sum x_i^2}{\sigma_0^2}\right) > F_{\chi_n^2}(\chi_{n, \alpha}^2) \right\}$$

$$= \left\{ \underline{x} : \alpha > 1 - F_{\chi_n^2}\left(\frac{\sum x_i^2}{\sigma_0^2}\right) \right\}$$

$$\therefore p(\underline{x}) = 1 - F_{\chi_n^2}\left(\frac{\sum x_i^2}{\sigma_0^2}\right)$$

Ex: $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$ Test $H_0: \sigma = \sigma_0$ vs. $H_1: \sigma \neq \sigma_0$. Using test fn. $\phi(\underline{x}) = \mathbb{1}\left(\frac{\sum x_i^2}{\sigma_0^2} > \chi_{n, \alpha/2}^2 \text{ or } < \chi_{n, 1-\alpha/2}^2\right)$

$$\therefore p(\underline{x}) = 2 \min \left\{ F_{\chi_n^2}\left(\frac{\sum x_i^2}{\sigma_0^2}\right), 1 - F_{\chi_n^2}\left(\frac{\sum x_i^2}{\sigma_0^2}\right) \right\}$$

Thm: Suppose $\{R_\alpha\}$ is a nested seq. of rejection regions for testing $H_0: \theta \in \Theta_0$ vs. $H_1: \theta \in \Theta_1$

① If $\sup_{\theta \in \Theta_0} \mathbb{P}_\theta(\underline{x} \in R_\alpha) \leq \alpha \quad \forall 0 < \alpha < 1$ (i.e., R_α = rejection region $p(\underline{x})$ stoch. larger than $U(0,1)$ of a level α test)
then for any $\theta \in \Theta_0$ $\mathbb{P}_\theta(p(\underline{x}) \leq u) \leq u \quad \forall 0 \leq u \leq 1$

② If for $\theta \in \Theta_0$, $\mathbb{P}_\theta(\underline{x} \in R_\alpha) = \alpha \quad \forall 0 < \alpha < 1$
(stronger than size- α condition)

then for any $\theta \in \Theta_0$, $\mathbb{P}_\theta(p(\underline{x}) \leq u) = u \quad \forall 0 \leq u \leq 1$

i.e. $p(\underline{x}) \sim U(0,1)$

Test: $\phi(\underline{x}) = \mathbb{1}(p(\underline{x}) \leq \alpha) \begin{cases} \longrightarrow \text{level-}\alpha \text{ in } \textcircled{1} \\ \longrightarrow \text{size-}\alpha \text{ in } \textcircled{2} \end{cases}$

Pf: ① $\{p(\underline{x}) \leq u\} = \{\underline{x} \in R_v \quad \forall v > u\}$
for any $\theta \in \Theta_0 \Rightarrow \mathbb{P}_\theta(p(\underline{x}) \leq u) \leq \mathbb{P}_\theta(\underline{x} \in R_v) \quad \forall v > u$

$$\Rightarrow \lim_{v \rightarrow u} \mathbb{P}_\theta(p(\underline{x}) \leq u) \leq \lim_{v \rightarrow u} v = u$$

$$\Rightarrow \mathbb{P}_\theta(p(\underline{x}) \leq u) \leq u \quad \text{—————} \textcircled{1}$$

$$② \{x \in R_u\} \Rightarrow \{p(x) \leq u\}$$

$$\mathbb{P}_\theta(p(X) \leq u) \geq \mathbb{P}_\theta(X \in R_u) = u \text{ ————— } ②$$

$$①, ② \Rightarrow \mathbb{P}_\theta(p(X) \leq u) = u \quad \begin{matrix} \forall \theta \in \Theta_0 \\ \forall u \in [0, 1] \end{matrix}$$

Qn When $\mathbb{P}_\theta(X \in R_\alpha) = \alpha \quad \forall \theta \in \Theta_0$ hold?

If $\Theta_0 = \{\theta_0\}$, have $\mathbb{P}_{\theta_0}(X \in R_\alpha) = \alpha$ for any size- α region R_α

$$\begin{matrix} X_1, \dots, X_{n_1} \stackrel{iid}{\sim} \mathcal{N}(\mu_1, \sigma^2) \\ Y_1, \dots, Y_{n_2} \stackrel{iid}{\sim} \mathcal{N}(\mu_2, \sigma^2) \end{matrix} > \text{indep.}$$

$$H_0: \mu_1 = \mu_2 \text{ vs. } H_1: \mu_1 \neq \mu_2$$

$$\Theta_0 = \{(\mu_1, \mu_2, \sigma^2) : \mu_1 = \mu_2 = \mu \text{ (say)}\}$$

$$R_\alpha = \left\{ \frac{|\bar{x} - \bar{y}|}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} > t_{n_1+n_2-2, \alpha/2} \right\}$$

$$\theta \in \Theta_0 \quad \mathbb{P}_\theta \left(\frac{|\bar{X} - \bar{Y}|}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} > t_{n_1+n_2-2, \alpha/2} \right) = \alpha$$

(Generalized) Likelihood Ratio Test (LRT):

$$X_1, \dots, X_n \sim f(x|\theta) : \theta \in \Theta \subseteq \mathbb{R}^k. \text{ Test: } H_0: \theta \in \Theta_0 \text{ vs.}$$

$$H_1: \theta \in \Theta_1$$

For simplicity, we consider $\Theta_0 \cup \Theta_1 = \Theta$

Intuitive development: N.P. Lemma $H_0: \theta = \theta_0$ vs. $H_1: \theta = \theta_1$

sf $\lambda(x) = \frac{L(\theta_1|x)}{L(\theta_0|x)}$ sufficiently large $\Rightarrow$ reject H_0

General: $\theta \in \Theta_0$ vs. $\theta \in \Theta_1$

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_1} L(\theta|x)}{\sup_{\theta \in \Theta_0} L(\theta|x)} = \frac{\text{max chance of getting } x \text{ when } H_1 \text{ true}}{\text{max chance of getting } x \text{ when } H_0 \text{ true}}$$

$\Rightarrow$ reject H_0 if suff. large

Equivalently

If $\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\sup_{\theta \in \Theta_1} L(\theta|x)} \in (0, \infty)$ suff. small $\Rightarrow$ reject H_0

$$\sup_{\theta \in \Theta} L(\theta|x) = \sup_{\theta \in \Theta_1} L(\theta|x) \left[\max \left\{ \sup_{\theta \in \Theta_0}, \sup_{\theta \in \Theta_1} \right\} \right]$$

Equivalently

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\sup_{\theta \in \Theta} L(\theta|x)} \in [0, 1]$$

suff. small $\Rightarrow$ reject H_0

So the LRT for $\theta \in \Theta_0$ vs. $\theta \in \Theta_1$, rejects H_0 if $\lambda(x) < c$ where c is s.t. $\sup_{\theta \in \Theta_0} P_{\theta}(\lambda(X) < c) = \alpha$

Comments: (1)

$$\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\sup_{\theta \in \Theta} L(\theta|x)} = \frac{\sup_{\theta \in \Theta_0} g_{\theta}(T(x))}{\sup_{\theta \in \Theta} g_{\theta}(T(x))}$$

So LRT is based on suff. stat $T(x)$

(2) Unlike UMP, no control over power (cannot say LRT has maximum power)

(3) If UMP or UMPU exists, often it coincides with the LRT test